

Dr. Nurdiana Z. Abidin

Senior Lecturer
DS13

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Researcher ID: O-3964-2016

Scopus ID: 56366272800

Google Scholar ID: C_jBuOYAAAAJ

Web of Science ID: HTM-3927-2023

Education

Ph.D in Biosciences
Oct 2016 - Feb 2021

University of Nottingham (MY, UK)
Main Supervisor: Assoc. Prof. Soma Mitra (School of Biosciences, Faculty of Science and Engineering, University of Nottingham Malaysia)
Dissertation: Osteosarcopenic Obesity: The Development of Screening Test Criteria and The Association with Bioavailable 25(OH)D

MSc. in Health Sciences (by research)
Nov 2012 - Apr 2015

Universiti Sains Malaysia
Main Supervisor: Professor Rabindarjeet Singh (Head of Lifestyle Science Cluster, Advanced Medical and Dental Institute, Universiti Sains Malaysia)
International Advisor: Professor Wendy Brown (Director, Centre for Research on Exercise, Physical Activity and Health, Professor of Physical Activity and Public Health, School of Human Movement and Nutrition Studies, The University of Queensland, Australia)
Dissertation: Physical Activity Levels of Malay Older Adults Objectively-Measured Using Tri-axial Accelerometer and Its Associated Factors

BSc. in Biomedical Sciences (Hons.)
Sep 2004 - June 2008

International Islamic University Malaysia
Main Supervisor: Assoc. Prof. Othman Abd Samah
Final Year Research Project: Antibacterial Activity of Ten Medicinal Plants Obtained From Some Selected Villages in the States of Kedah and Penang, Malaysia.

Certificates, Publications & Presentations

Certificates

- Good Clinical Practice (*6 – 8 May 2010, Universiti Sains Malaysia, Penang, Malaysia*)

Publications (JCR & Citation Indexed Journal)

[1] Abidin NZ, Mansor NS, Yusof HA, Abd Hamid AI, Mat Isa S. 2026. The Impact of Dual-Task Training Versus Single-Task Training on Cognitive and Physical Function in Neurologically Impaired Individuals: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. Percept Mot Skills. 2026. OnlineFirst. https://doi.org/10.1177/00315125261425126

[2] Wen F, Yusof HA, **Abidin NZ**. 2025. Effects of Active Video Games on Energy Expenditure and Weight Management Among Young Adults: A Systematic Review. Games Health J. 2025;14(5):345–357. https://doi.org/10.1089/g4h.2025.0044

[3] Li MZ, Zamri E, Liu K, **Abidin NZ**, Mei X. 2025. Fall occurrence and association with knowledge and practice on fall prevention among elderly residents in nursing homes, PR China. Southeast Asian J Trop Med Public Health. 2025;56(4):489–517.

[4] Wen F, Yusof HA, **Abidin NZ**. 2025. Effects of active video games on affective responses and mental health symptoms in young adults: a systematic review. BMC Public Health. 2025;25:3995. https://doi.org/10.1186/s12889-025-25413-1

[5] Abidin NZ, Ooi CH, Nosaka K, et al. 2024. Effects of Resveratrol Supplementation on Delayed Onset Muscle Soreness and Muscle Recovery: A Systematic Review. Malays J Med Sci. 2024;31(6):77–102. https://doi.org/10.21315/mjms2024.31.6.7

[6] Abidin NZ. 2023. The "Fat or Fit Paradox" in postmenopausal women: A cross-sectional study on the association between skeletal muscle mass, adiposity, and muscle strength. Muscle, Ligaments and Tendons Journal, 13(3), 494-503. https://doi.org/10.32098/mltj.03.2023.20

[7] Abidin NZ. 2023. Comparison of muscle, bone and fat indices between stages of sarcopenia in postmenopausal Malaysian women. Malays J Med Sci. 2023;30(5):91–105. https://doi.org/10.21315/mjms2023.30.5.8

[8] Abidin NZ. 2023. Obesity and Musculoskeletal Health of Young and Older Malaysian Women: A Cross-Sectional Study. Abidin NZ. Obesity and musculoskeletal health of young and older Malaysian women: a crosssectional study. Malays J Med Sci. 2023;30(1):137–151. https://doi.org/10.21315/mjms2023.30.1.12

[9] Abidin NZ, Mitra SR. 2022. [Abstract] Osteosarcopenic Obesity: The Interrelationship Between Fat, Bone And Muscle and their Associations to Total and Bioavailable 25(Oh)D Levels in Postmenopausal Women. *The Journal of Frailty & Aging,* **11,** 46–82. https://doi.org/10.14283/jfa.2022.23

[10] Mitra SR, **Abidin NZ.** 2021. Comparison of anthropometric profile, bone-related parameters, biochemical parameters, functional capacity and Vitamin D status - a study on postmenopausal Malaysian women. *Malaysian Journal of Nutrition*. [doi: https://doi.org/10.31246/mjn-2020-0080]

Certificates, Publications & Presentations (cont.)

[11] Abidin NZ, Mitra SR. 2021. Determination of cut-off values for the screening of Osteosarcopenia in obese postmenopausal women. *Current Gerontology and Geriatrics Research*, vol. 2021, Article ID 6634474, 15 pages, 2021. https://doi.org/10.1155/2021/6634474

[12] Abidin NZ, Mitra SR. 2021. Total vs. Bioavailable: Determining a Better 25(OH)D Index in Association with Bone Density and Muscle Mass in Postmenopausal Women. *Metabolites,* 11(1), 23; https://doi.org/10.3390/metabo11010023

[13] Abidin NZ, Brown W, Clark B, Muhamed AMC, Singh R. 2016. Physical Activity Measurement by Accelerometry among Older Malay Adults Living in Semi-rural Areas – Feasibility Study. *Journal of Aging and Physical Activity,* 24(4):533-539. Epub 2016 Aug 24.

[14] Abidin NZ, Brown W, Clark B, Muhamed AMC, Singh R. 2016. Assessing The Physical Activity Level of Elderly Malays Living in Semi-Rural Areas Using the Tri-axial Accelerometer. *Malaysian Journal of Nutrition*, Vol. 22 Issue 3, p363-374. 12p.

[15] Abidin NZ, Muhamed AMC, Singh R. 2016. Associations between Tri-Axial Accelerometer-Determined Physical Activity and Self-Reported Heath-Related Quality Of Life of Older Malay Adults Using Cut-Points for Vector Magnitude. *Malaysian Journal of Movement, Health & Exercise,* [S.l.], v. 5, n. 1, aug. 2016. ISSN 2600-9404. doi:https://doi.org/10.15282/mohe.v5i1.102

[16] Abidin NZ, Muhamed AMC, Singh R. 2014. Physical Activity Level and Sedentary Behavior of Malay Adults Objectively Measured Using a Tri-Axial Accelerometer: A Preliminary Study. *Asian Journal of Exercise & Sports Science*, 11 (2); 56-66.

[17] Mohamed NG, **Abidin NZ**, Law KS, Abe M, Suzuki M, Muhamed AMC and Singh R. 2014. The effect of wearing sanitary napkins of different thicknesses on physiological and psychological responses in Muslim females. *Journal of Physiological Anthropology*, 33 (28), doi:10.1186/1880-6805-33-28

[18] Samah OA and **Abidin NZ**. 2009. Antibacterial Activity of Ten Medicinal Plants Obtained from some Selected Villages in the States of Kedah and Penang, Malaysia. *Indonesian Journal of Pharmacy*, 20 (2); 99-103

Conference books and articles

[1] AMDI Newsletter: **Abidin NZ**. 4th June 2025. Mechanisms of Augmented Reality (AR) and Virtual Reality (VR) on the Human Brain. Newsletter. 2025:1–7.

[2] AMDI Newsletter: **Abidin NZ**. 2nd July 2024. Innovative Healthcare: The Rise of Augmented Reality (AR) and Virtual Reality (VR). Newsletter. 2024.

[3] AMDI Newsletter: **Abidin NZ**. 10th June 2024. Virtual Fitness: Revolutionizing Health and Wellness Through Immersive Exercise. Link: https://news.amdi.usm.my/fullnews.php?id=eGVTSy9mZnRMZzJoNCs2QXVFVlZhZz09

[4] AMDI Newsletter: **Abidin NZ**. 2nd July 2024. Innovative Healthcare: The Rise of Augmented Reality (AR) and Virtual Reality (VR). Link: https://news.amdi.usm.my/fullnews.php?id=cVJQbEgzQjRPRUIvSExycUlKQ2RVZz09

[5] AMDI Newsletter: **Abidin NZ**. 23rd August 2022. The Obesity Paradox: Could Obesity Be Beneficial to You? Link: https://news.amdi.usm.my/fullnews.php?id=dG9qeXJNZmJHelNqM1dYOUJEZzA4UT09

[6] AMDI Newsletter: **Abidin NZ**, Mitra SR. 5th of July 2021. Total vs. Bioavailable: Determining a Better 25(OH)D Index in Association with Bone Density and Muscle Mass in Postmenopausal Women. Link: https://news.amdi.usm.my/fullnews.php?id=dW8ydU9FLzVkNzBsb0ptV0tYZnFidz09

[7] Abstract: **Abidin NZ**, Muhamed AMC, Singh R. 2013. Preliminary Observation on Objective Measurement of Physical Activity Level and Sedentary Behaviour using Tri-axial Accelerometer in Malay Older Adults.
International Sports Science Student Conference 2013, at University of Malaya, Kuala Lumpur, Malaysia, 28th & 29th November 2013

[8] Abstract: Singh R, **Abidin NZ**, Mohamed NG, Ismail N, Ahmad N. 2011. Hydration status and Fluid Balance of Players during Real-Time Soccer Matches Played during the Malaysian League.
4th AFC Conference on Science and Football Medicine, Kuala Lumpur, 18th to 20th March 2011

[9] Abstract: Samah OA and **Abidin NZ**. 2009. Screening of Antibacterial Activity of Ten Medicinal Plants Obtained from some Selected Villages in the States of Kedah and Penang, Malaysia.
The 3rd Regional Conference on Natural Resources in the Tropics: Harnessing Tropical Natural Resources through Innovations and Technologies 2009, Hilton Kuching, Sarawak, Malaysia, 3rd to 5th August 2009

[10] Magazine Article: **Abidin NZ.** Physical Activity in Older Adults
Lifestyle Science Cluster Wellness Bulletin: 4th Issue 2011

[11] Magazine Article: **Abidin NZ**. Eat Your Way to Weight Loss
Lifestyle Science Cluster Wellness Bulletin: 3rd Issue 2010

Presentations

Oral presentations

[1] Abidin NZ, Mitra SR, Ho WK. 2018. Development of screening test criteria for osteosarcopenia in obese postmenopausal Malaysian women. *Nottingham-Adelaide Postgraduate Symposium 2018*, 11 October 2018, The University of Nottingham in Malaysia

[2] Abidin NZ, Brown WJ, Clark B, Muhamed AMC, Singh R. 2015. Associations between Accelerometer-determined Physical Activity and Self-reported Heath-related Quality of Life in Semi-rural Older Malay Adults. *International Conference on Movement, Health and Exercise 2015*, 5th – 7th October 2015, The Royale Bintang Hotel, Penang, Malaysia

[3] Abidin NZ, Muhamed AMC, Singh R. 2013. Preliminary Observation on Objective Measurement of Physical Activity Level and Sedentary Behaviour using Tri-axial Accelerometer in Malay Older Adults. *International Sports Science Student Conference 2013*, 28th & 29th November 2013, University of Malaya, Kuala Lumpur, Malaysia

Certificates, Publications & Presentations (cont.)

[4] Abidin NZ, Muhamed AMC, Singh R. 2013. Are Malaysian Elders Active? Objective Assessment of Physical Activity Level in Community-Dwelling Older Adults. *Universiti Sains Malaysia -The University of Sydney -The University of Queensland Joint Research Seminar in Exercise, Nutrition, Health And Wellness*, 25th – 28th February 2013, The University of Sydney, Australia and The University of Queensland, Australia

Poster presentations

[1] Abstract: **Abidin NZ**, Mitra SR. 2022. Osteosarcopenic Obesity: The Interrelationship Between Fat, Bone And Muscle and their Associations to Total and Bioavailable 25(Oh)D Levels in Postmenopausal Women. International Conference on Frailty and Sarcopenia Research, April 20-22, 2022 Boston, MA, USA

[2] Abstract: **Abidin NZ**, Tay HX, Mitra SR. 2018. The Influence of Body Fat on the Association of Serum 25-hydroxyvitamin D and Muscle Strength in Postmenopausal Malaysian Women. *33rd Annual Scientific Conference of Nutrition Society of Malaysia*, 8 - 10 May 2018, Hotel Istana Kuala Lumpur

[3] Abstract: Mitra SR, **Abidin NZ**. 2017. Musculoskeletal health in postmenopausal Malaysian Malay women. *10th Asia Pacific Conference on Clinical Nutrition*, 26 - 29 November 2017, Adelaide Convention Centre, Australia

[4] Abstract: Abidin NZ, Muhamed AMC, Singh R. 2014. How Active Are Our Malay Elderly Population? *1st Postgraduates Colloquium on Translational Research 2014*, 9th & 10th September 2014, at Animal Research Complex, Advanced Medical and Dental Institute, Universiti Sains Malaysia

Book Chapter

[1] Book Title: Genetic Disorders & Rare Diseases: Current Updates
Chapter Title: Osteosarcopenic Obesity: A Comprehensive Guide to Prevention and Management
Publisher: Nova Science Publishers, Inc., USA
Citation: Shaharudin, S., Edinur, H. A., & Che Mat, N. F. (2023). Genetic Disorders and Rare Diseases: Current Updates. School of Health Sciences, Universiti Sains Malaysia.
https://doi.org/10.52305/WNGG1520

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[1] Cellular Lifespan Part I_CRFM2023P02711 (Film)

[2] Cellular Lifespan Part II_ CRM2024 P03804 (Film)

[3] Cellular Lifespan Part II_ CRFM2025P07518 (Film)

[4] DADU DODECA-FIT_ CRAR2025P07523 (Art)

Key Research Projects

Principal Investigator Dr. Nurdiana Z. Abidin 2022 – present	**Fundamental Research Grant Scheme** Title of Research: Mechanistic evaluation of resveratrol-loaded dissolvable microneedle patch and its effects on delayed onset muscle soreness induced by exercise
Principal Investigator Dr. Nurdiana Z. Abidin 2022 – present	**Dementia Multidisciplinary Research-IPPT** Title of Research: The Effects of Dual-Task Cognitive Training in Fully Immersive Virtual Reality on Cognitive Function, Brain Functional Connectivity and Physical Performance in Older Adults with Mild Cognitive Impairments: Feasibility Study and Randomized Controlled Trial
Principal Investigator Professor Rabindarjeet Singh 2011 - 2013	**Longitudinal Study on the Lifestyle Intervention of an Adopted Community** Title of Research: Effectiveness of a Healthy Lifestyle Program in an Adopted Community
Principal Investigator (MY) Professor Rabindarjeet Singh 2011 – 2013	**Research collaboration with KAO Corporation, Japan** Title of Research: Effects of Sanitary Napkins on Physiological Stress Markers and Quality of Life of Malaysian Women
Principal Investigator Professor Rabindarjeet Singh 2010 – 2011	**Hydration Study on Football Players** Title of Research: Hydration Status and Fluid Balance of Players during Real-time Soccer Matches Played during the Malaysia League
Principal Investigator Dr. Ahmad Munir Che Muhamed 2010 – 2011	**Caloric Restriction on Cognitive Tasks** Title of Research: The Effect of Caloric Restriction during Fasting on Cognitive Tasks

Grants Obtained

[1] Year: 2024 - present
Funder: Short Term Grant Universiti Sains Malaysia
Amount: RM 45,070.00

[2] Year: 2022 - present
Funder: Fundamental Research Grant Scheme (FRGS). Ministry of Higher Education Malaysia
Amount: RM 171,800

[3] Year: 2012 - 2014
Funder: Advanced Medical and Dental Institute Postgraduate Research Student Fund, Universiti Sains Malaysia
Amount: RM 40,000

Grants Obtained (cont.)

[4] Year: 2012 - 2014
Funder: Postgraduate Research Scheme Grant from the Institute of Postgraduate Study, Universiti Sains Malaysia
Amount: RM 10,000

Membership of Professional Organizations

- The Physiological Society UK
- Nutrition Society of Malaysia (NSM)
- Malaysia Healthy Ageing Society

Fellowships, Awards, and Honors

- Universiti Sains Malaysia Fellowship under the Academic Staff Training Scheme (ASTS) : 2012-2021
- Academy Fiesta@USM 2025 – *Silver in Micro-credential category* : 2025
- USM–AUSC International Sports Innovation Challenge 2025 *Gold Medal and First Runner Up in Special Category* : 2025

Courses Taught

Micro-credential Course Developer (9 videos)

Module 1: Foundations To Exercise Physiology
Module 5: Exercise Physiology for Special Population

TCE 504 Applied Exercise Physiology
(*Advanced Medical and Dental Institute, Universiti Sains Malaysia*)

- Cellular Lifespan
- Exercise Immunology
- Age-related Physiological Changes
- Cardiorespiratory and Neuromuscular Responses to Exercise of Aging Population

TCE 515 Clinical Exercise Physiology [Head of Module]
(*Advanced Medical and Dental Institute, Universiti Sains Malaysia*)

- Exercise Physiology for Neurological Disorders
- Exercise Physiology for Bone and Joint Disorders

Courses Taught (cont.)

TCE517 Practicum
(Advanced Medical and Dental Institute, Universiti Sains Malaysia)

- Supervisor

TMR 507 Research
(Advanced Medical and Dental Institute, Universiti Sains Malaysia)

- Internal examiner

Digital Innovation / Educational Technology Development

[1] PRIME-Active: A Digital Health Platform for Active Ageing and Community-Based Physical Activity Promotion

Principal Developer | 2026–present

Developing a prototype digital health platform designed to support physical activity engagement, self-monitoring, and health promotion among older adults and community populations. The platform is being developed as part of a community-focused digital health innovation initiative and will be further refined through stakeholder input, co-design processes, and pilot implementation.

Status: Prototype under development
Link: Not publicly available at this stage

[2] PhysioVerse 3D: Interactive Biomechanics Platform

Founder / Developer | 2026

Web-based interactive 3D educational platform for exercise physiology and biomechanics

Developed an interactive digital learning platform designed to enhance the teaching and understanding of exercise biomechanics, movement analysis, and muscle activation patterns. The platform uses animated 3D models to demonstrate selected exercises, supported by structured educational content on primary and secondary muscles, joint actions, movement mechanics, common errors, coaching cues, and clinical relevance.

The platform was developed to support visual and self-directed learning among students in Clinical Exercise Science and related health science disciplines, particularly in bridging theoretical knowledge with applied movement analysis.

Key features:

- Interactive 3D exercise demonstrations
- Exercise-specific muscle activation mapping
- Biomechanics and joint movement explanations
- Clinical and educational notes for applied exercise physiology
- Web-based access for flexible student learning

Link: https://physio-verse-3-d.vercel.app

Digital Innovation / Educational Technology Development (cont.)

[2] MSc Clinical Exercise Science Quiz App

Founder / Developer | 2026

Mobile-accessible digital quiz platform for postgraduate Clinical Exercise Science learning

Designed and developed a digital quiz application to support MSc Clinical Exercise Science students in revising key concepts across core modules. The app provides structured practice questions to encourage frequent revision, terminology familiarisation, and self-directed learning outside formal classroom hours.

The platform was created as a teaching innovation to improve student engagement, reinforce applied clinical knowledge, and allow students to monitor their own learning progress independently.

Key features:

- Module-based question organisation
- Multiple-choice practice questions
- Immediate answer feedback
- Localised progress tracking visible only to the user
- Web and mobile-accessible learning format
- Designed to support repeated practice and knowledge reinforcement

Link: https://clinical-quiz-ces.vercel.app/

Leadership/Administrative Experiences

- Head of Program for MSc Clinical Exercise Science (Coursework) 2023/2024, 2024/2025, 2025/2026
- Project Leader/Chairman for:

  [1] Karnival Warga Emas 2022
  In collaboration with Ministry of Higher Education Malaysia and Masjid Abdullah Fahim
  12th November 2022, Masjid Abdullah Fahim Bertam, Kepala Batas
- Founder of AMDI Longitudinal Program for Healthy Ageing (ALPHA)
- Established, managed, and maintained a state-of-the-art laboratory, diligently overseeing inventory and record-keeping for essential supplies and consumables
- Spearheaded the organization and execution of numerous national and international conferences and seminars, engaging interdisciplinary staff from the Advanced Medical and Dental Institute, public communities, and private sectors under the auspices of the Lifestyle Science Cluster
- Facilitated and coordinated a large-scale, impactful community project assessing the effectiveness of a Healthy Lifestyle Program within a designated community, demonstrating commitment to translating research into real-world applications

Leadership/Administrative Experiences (cont.)

- Chief Secretary of the following:

  [1] International Fasting and Sustainable Health Conference: The Science of Fasting 2010
  20th – 21st December 2010, Eastin Hotel, Queensbay, Penang, Malaysia
  [2] Malaysia-Australia Research Colloquium on Exercise, Nutrition, Health and Wellness 2011
  19th – 21st September 2011, Park Royal Resort, Batu Feringghi, Penang, Malaysia
  [3] International Conference on Movement, Health and Exercise 2015 (MoHE2015)
  5th – 7th October 2015, The Royale Bintang Hotel, Penang, Malaysia

Supervisory Experiences

[1] Main supervisor
Name of student: Tie Yuhe
(Master of Science (Research - Full Time) at Universiti Sains Malaysia) 2024
The Therapeutic Potential of 24-style Tai Chi in Modulating Type A Personality Behavior in Primary School Students

[2] Main supervisor
Name of student: Huang Weijie
(Doctor of Philosophy (Research - Full Time) at Universiti Sains Malaysia) 2022
Cultural Influences on Mental Training Techniques in Basketball: A Comparative Analysis of Eastern and Western Approaches

[3] Co-supervisor
Name of student: Zhang Di
Master (Research - Full Time) at Universiti Sains Malaysia 2023
Effect Combination of Conventional Rehabilitation Therapy with sling Exercise Therapy Training on Early Recovery of Knee Joint Function after Meniscus Repair

[4] Co-supervisor
Name of student: Nazirah Binti Gulam Mohamed
(Doctor of Philosophy (Research - Full Time) at Universiti Sains Malaysia 2022
Effects of Eccentric Exercise On Weight Loss, Physiological and Psychological Adaptations Among Obese Women: A Community Based Approach

[5] Co-supervisor
Name of student: Wen Feng
Doctor of Philosophy (Research - Full Time) at Universiti Sains Malaysia 2022
The Study On Effects Of Combination In Different Exercise Methods And Dietary Habits Of College Students On Physical Health In China

[6] Co-supervisor
Name of student: Li Muzheng
Master (Research - Full Time) at Universiti Sains Malaysia 2022
Assesment of Balance Measurement, Knowledge and Practice of Falls Prevention Among Elderly in the Nursing Homes

Services to the University/Committee

[1] Head of Program (MSc Clinical Exercise Science 2023/2024, 2024/2025, 2025/2026)

[2] Academic Quality Committee

[3] Academic Accreditation Committee

[4] Technology Driven Education Committee

[5] Academic Program Promotion Committee

[6] Academic Advisor

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